

JotForm Widgets - Create your own form field

JotForm Widgets - Create your own form field

What are JotForm Widgets?

[Jotform widgets](#) are customizable form fields that can be added to JotForm forms. JotForm widgets extend regular form fields which take an input and save it on the form. When a widget is created, it can be programmed to do anything. Like the [Comparison Slider Widget](#), they can be used to simply show dynamic content on forms. Alternatively, like the [Annotate picture widget](#), they can interact with the user and save the input when the form is submitted.

Here are some of the creative ways a widget can be used on JotForm forms:

1. Perform simple or complex calculations using their form -> [browse examples](#)
2. Play a video or share a video stream -> [browse examples](#)
3. Creatively edit and save text -> [browse examples](#)
4. Upload and save pictures, videos and other files -> [browse examples](#)

Developing a custom widget requires a good understanding of the JotForm Widgets API accessible via `JFCustomWidget()`.

Create your own JotForm Widget

This is a step by step tutorial on how to create a custom widget that can be used on JotForm forms. We will create a very simple widget. It will add a text input on the form. The text input will be configured. Here is what the end result will look like:

<https://form.jotform.com/201001774864047>

Here is the code to the example above:

```
<!DOCTYPE html>
<head>
  <meta charset="utf-8">
  <script src="//js.jotform.com/JotFormCustomWidget.min.js"></script>
</head>
<body>
  <div id="mywidget">
    <input type="text" id="myTextInput">
  </div>
  <script type="text/javascript">
    /*Begin by subscribing to the "ready" event.
```

```

The ready event has a callback function
where you will include all your JavaScript. */
JFCustomWidget.subscribe("ready", function(){
    /*Next, get Widget's user settings to apply them on your widget.
    In this case, we are setting the maxLength attribute for our
input field*/
    var charLimit =
parseInt(JFCustomWidget.getWidgetSetting('charLimit'));
    var myTextInput = document.getElementById("myTextInput");
    myTextInput ? myTextInput.maxLength = charLimit : "";
    //Then subscribe to form submit event to save the input when the
form is submitted
    JFCustomWidget.subscribe("submit", function(){
        /*prepare the data you'll be sending
        setting "valid" as "true" enables the form to check for a
value if it is required*/
        var data = {
            valid: true,
            value: myTextInput.value
        }
        // Finally, send data to your JotForm form
        JFCustomWidget.sendSubmit(data);
    });
});
</script>

</body>
</html>

```

Steps

1. To begin, copy the code above to an html file, save it and upload it to a location of your choice.
2. Go to the [widgets landing page](#), and click on **Add New Widget** at the bottom right of the page.


 A rectangular button with a light green border. On the left, there is a small green square containing a white plus sign. To the right of the plus sign, the text "Add New Widget" is written in a dark green, sans-serif font.

3. This will open a form, which you will fill and submit. Once submitted, the widget will be available on your Account from the JotForm Builder, and you should be able to test it using any of your forms. The following describes the information required on the form:
 - **Widget Icon** - this icon will identify your widget on the [widget directory](#) and on the form builder.
 - **Widget Demo Screenshots** - take screen captures of your widget in action. This will be shown on the [widget directory](#) and the form builder.

- **Name** - A unique name that will be used to identify your widget from all others - while concisely indicating what your widget does.
- **Short Description** - A concise description or benefit of what your widget does.
- **Description** - Describe your widget function, and the value it will provide to users. You may add some details about its configuration if any. Any third party services that require attribution should be linked here as well.
- **Demo URL** - Optional - Add a URL with a form which shows your widget in action.
- **Video URL** - Optional - A video that shows how the widget works, or shows off the different features and benefits of your widget.
- **Categories** - Pick at least one category that best defines your widget function.
- **Widget Type** - The only option available publicly is "Iframe Widget".
- **Widget Width** - the Iframe width in pixels. This can be adjusted programatically via `setFrameSize`.
- **Widget iFrame URL** - This is the secure (https://...) url to your html or php file that loads the page and scripts you have created for your widget. Your widget's user interface appears on this URL. The URL will be loaded in an iFrame.
- **Secure Widget** - If your form submits data to a third party source other than the form, do not check this. This helps us to align third party services to our [privacy_policy](#) and protect user information.
- **Static Widget** - If you check this, it means that your widget does not send any data to the form, the widget just displays content in the iframe.
- **Multiple Instance** - If you check this, your widget can be added multiple times on the form. Some usecases restrict using a form field more than once on a form, so this is a useful tool should you need it.
- **Required by default** - When the widget is added on the form, it must have a input value before the forms is submitted. The user can always turn off this setting from the field settings in the JotForm builder.
- **Default Label** -Optional - This is the default label when your widget is added on the form
- **Question Label** - Some form fields do not require a question label. Leave this unchecked if you don't need one.
- **Label alignment** - this is the default position of the question label if your widget has one.
- **User Parameters** - These are settings which are configured via the JotForm form builder for your widget. This provides a way for form creators to customize your widget for specific scenarios. You can access these settings via `getWidgetSetting` and `getWidgetSettings`. Clicking on *Add Parameter* will open a form which will require the configuration below. When you have two or more user parameters all set up, you can drag them into position if you need to reorder them for the user interface on the JotForm form builder. Here is what is required to set up a parameter:

- **Name** - a camelCase name which will be accessible via `getWidgetSetting` and `getWidgetSettings`.
- **Human Readable Name** - this is what will be shown on the JotForm form builder widget settings interface.
- **Field Type** - These are the field types which will be used to collect your settings from the widget settings interface on the JotForm form builder. Select a field type depending on what you need - here are your options:
 - **Color** - This shows a color picker on the JotForm builder widget settings. The color value is saved as `hex` or `rgba`.
 - **Text** - This shows a short text input field on the JotForm builder widget settings.
 - **Textarea** - This shows a long text input field on the JotForm builder widget settings.
 - **Dropdown** - This shows a dropdown field on the JotForm builder widget settings.
 - **Radio** - This shows a radio button field on the JotForm builder widget settings.
 - **Checkbox** - This shows a checkbox button on the JotForm builder widget settings.
 - **Secure Text** - This shows a text input field on the JotForm builder widget settings. The user input is encrypted, and can only be used when decoded on a backend script. It is useful for authorization tokens, or any sensitive text that requires protection. Please refer to [Security considerations for your widget](#) to learn how to use it.
 - **Multiple Image** - An image uploader interface on the widget user settings. Useful if your widget needs images from the widget settings.
 - **Date** - A date picker is shown on the JotForm builder widget settings. The date format is `yyyy-mm-dd`.
 - **External** - This opens a pop-up window with a custom URL. It's useful for authorization flows, or calling endpoints to fetch settings on a third party location using a custom UI. Please refer to [Security considerations for your widget](#) to learn how to use it.
 - **Remote Options** - This field requires presets to fetch settings or options from a third party location. The options are populated in a dropdown in the JotForm builder widget settings.
 - **Schema** - to be determined.
 - **Api Auth** - This requests authorization to access to JotForm API. Please refer to [Security considerations for your widget](#) to learn how to use it.
 - **Key Value Pair** - to be determined
 - **File Upload** - This shows a file upload interface on the JotForm builder widget settings.

Security considerations for your widget

JotForm Widgets API Reference

Your widget will operate like a single page application, with an interactive user interface and ways of sending information to and from the form. There are three objects which will provide all the tools, properties and methods you need to interact with a JotForm form.

JFCustomWidget

1. `clearFields`
2. `clearWidgetFields`
3. `detectFrameMovement`
4. `getAllQueryString`
5. `getFieldsValueById`
6. `getFieldsValueByName`
7. `getFormTrackerID`
8. `getFrameData`
9. `getParentOrigin`
10. `getQueryString`
11. `getWidgetData`
12. `getWidgetSetting`
13. `getWidgetSettings`
14. `hideWidgetContainer`
15. `hideWidgetError`
16. `isFromCardform`
17. `isWidgetOnBuilder`
18. `isWidgetRequired`
19. `isWidgetStatic`
20. `listenFromField`
21. `listenFromWidget`
22. `makeWidgetNotRequired`
23. `makeWidgetRequired`
24. `off`
25. `on`
26. `publish`
27. `questionID`
28. `removeSubmittedFrame`
29. `replaceWidget`

30. `requestFrameResize`
31. `sendData`
32. `sendReady`
33. `sendSubmit`
34. `sendTranslatables`
35. `setFieldsValueById`
36. `setFieldsValueByLabel`
37. `setFrameSize`
38. `setFrameStyles`
39. `setupLibraries`
40. `showWidgetContainer`
41. `showWidgetError`
42. `storeToField`
43. `subscribe`
44. `texts`
45. `unsubscribe`

JFCustomWidgetUtils

1. `animateUntil`
2. `buildMetadata`
3. `decodeHtmlEntities`
4. `domReady`
5. `encodeHtmlEntities`
6. `executeUntil`
7. `extractMetadata`
8. `fixS3UriOrigin`
9. `getFormDomain`
10. `getIE`
11. `getS3UriRelativePath`
12. `hasMetadata`
13. `isArray`
14. `isEmpty`
15. `isFirefox`
16. `isJsonString`

17. `isMobile`
18. `isObject`
19. `isSafari`
20. `loadCSS`
21. `loadJS`
22. `resetS3UriOrigin`

JFWidgetTranslation

1. `getDictionary`
2. `getTranslatable`
3. `getTranslation`
4. `setDictionary`
5. `translate`